Business Question and Visualization Report

Date	24 September 2024
Team ID	[Enter Team ID]
Project Name	Global Fuel Prices and Taxation Analysis (2020 to 2026)
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

1. **Which countries tax fuel the most, and how does tax differ by region?**
    - *Visualization:* Filled map (1 TAX MAP) colored by average tax percentage.
    - *Insight:* Oceania has the highest average tax (38.7%) and Europe is next (34.6%). South America is lowest at 18.2%.
    - *[Insert screenshot of the Tableau sheet for question 1]*
2. **How does average tax compare across regions and subsidy levels?**
    - *Visualization:* Bar chart (2 Tax vs Subsidy) of average tax by region, colored by subsidy level.
    - *Insight:* Tax is similar across subsidy levels within a region (about 15% for High, Medium and Very High, 36% to 39% for Low), so low-subsidy countries tax more.
    - *[Insert screenshot of the Tableau sheet for question 2]*
3. **How do subsidy levels affect average fuel price in each region?**
    - *Visualization:* Bar chart (3 Subsidy vs Price) of average fuel price by subsidy level and region.

- *Insight:* Average fuel price falls from about $3.09 per liter at Low subsidy to $0.63 at High and $0.13 at Very High.
- *[Insert screenshot of the Tableau sheet for question 3]*

4. **How have diesel prices changed over time, and what is expected next?**
   - *Visualization:* Line chart with a 12-month forecast (4 Diesel Forecast).
   - *Insight:* The yearly average diesel price rose from $1.39 per liter in 2020 to $2.47 in 2026, and the forecast continues the upward trend.
   - *[Insert screenshot of the Tableau sheet for question 4]*
5. **Which 10 countries have the highest average diesel price?**
   - *Visualization:* Horizontal bar chart (5 Top 10 Countries), colored by region.
   - *Insight:* Hong Kong ($5.10), Ireland ($4.55) and Norway ($4.43) lead; most of the top 10 are in Europe.
   - *[Insert screenshot of the Tableau sheet for question 5]*
6. **Which 10 countries have the lowest average diesel price?**
   - *Visualization:* Horizontal bar chart (5b Bottom 10 Countries), colored by region.
   - *Insight:* Venezuela ($0.02), Iran ($0.03) and Libya ($0.05) are cheapest; most are oil-producing countries in the Middle East and Africa.
   - *[Insert screenshot of the Tableau sheet for question 6]*
7. **Which regions have the most volatile diesel prices each year?**
   - *Visualization:* Heatmap (6 Volatility Heatmap) of diesel volatility by region and year.
   - *Insight:* The Middle East is the most volatile in every year (about 1.2), while Oceania is the steadiest (0.06 to 0.20).
   - *[Insert screenshot of the Tableau sheet for question 7]*
8. **How does average fuel tax differ across income levels?**
   - *Visualization:* Bar chart (7 Tax by Income) colored by income level.
   - *Insight:* High-income countries tax fuel at 34.7%, against 22.9% for low-income and 21.1% for middle-income countries.

- *[Insert screenshot of the Tableau sheet for question 8]*

9. **How is subsidy spread across income levels?**

   - *Visualization:* 100% stacked bar chart (8 Subsidy by Income) of subsidy level share within each income level.

   - *Insight:* 87.5% of high-income records have Low subsidy, while low-income countries have a mix: 34.8% Low, 30.4% Medium, 17.4% High and 17.4% Very High.

   - *[Insert screenshot of the Tableau sheet for question 9]*

10. **How do petrol, diesel and LPG prices differ by income level?**

    - *Visualization:* Highlight table (9 Price by Income) of average price per fuel type by income level.

    - *Insight:* High-income countries pay about $3.55 per liter for petrol, against $1.39 in low-income and $1.59 in middle-income countries; diesel and LPG follow the same order.

    - *[Insert screenshot of the Tableau sheet for question 10]*